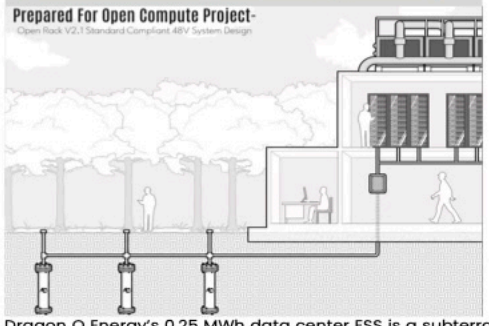
DATA CENTER 48VDC ESS - CONCEPT

Efficient and Fire Resistant Energy Storage



DRAGON Q ENERGY

Dragon Q Energy's data center ESS concept delivers safe, robust, and efficient, large-format energy storage where space is constrained and real estate is at a premium.

Dragon Q Energy's 0.25 MWh data center ESS is a subterranean battery system engineered for mission-critical, space-constrained sites where uptime and safety are non-negotiable. Installed fully below ground, it delivers high-density, large-format storage without occupying surface area and remains naturally shielded from environmental and human risks.

Designed for high-heat and high-humidity conditions, the system operates within a hermetically sealed, Argon-filled environment where fire cannot occur. Its corrosion-resistant construction and passive geothermal cooling further enhance efficiency and durability. Once buried, the ESS becomes a low-maintenance, long-life asset—providing stable, reliable power tailored to data center demands. Furthermore, the system incoperates 48VDC PSU topology for new-build data centers.

Value Propositions:

- **Maximize Usable Space –** Unlock underground capacity to add 0.25 MWh of energy storage per unit where real estate is constrained- leaving surface available for further use.
- **Ensures Operational Safety –**In addition to five on-board safety mechanisms designed to vent and prevent the spread of thermal runaway, the subterranean, fire-resistant design eliminates exposure risks for both personnel and critical assets.
- **Optimized Power Usage Effectiveness –** By integrating passive geothermal cooling and hermetically sealed underground architecture, the system minimizes auxiliary energy demand for thermal management. This improves overall PUE, delivering more usable power to IT loads while reducing both operational costs and environmental footprint.
- **Standards & Hyperscaler Alignment** – Designed in line with hyperscaler (META/Google) modernization efforts and the Open Compute Project's 48VDC Rack & Power initiatives, ensuring compatibility with next-generation data center infrastructure.

www.dragonqenergy.com | info@dragonqenergy.com | 805-421-6888

Proposed Technical Specifications

Performance Specifications

Chemistry	Li- Ion (NMC)
DC Voltage - 14S (Nominal)	48 VDC (51.8 V Nominal)
Cell Type	46120
Cell Capacity (Ah)	42 Ah
Cell Voltage (Nominal)	3.7V
Cell Energy	155Wh
Cell Configuration (Data Center)	127P14S (1778 Cells)
Total Pack Energy	.25 MWh
Pack Capacity (Ah)	5,334 Ah
Moderate Charge/Discharge Rate	2,500 A (0.5C)
Power (W)	129.5 KW
Internal Resistance (IR)	0.55 mΩ per pack
Ripple Volate	< 0.5% @ 2500A
Temperatuture Regulation	Passive Geothermal
System Round Trip efficiency	0.99
Battery Management System (BMS)	Active Balancing, RS485, CAN Bus

Environmental Specifications

Pack Operating Temperature (Max Permissiable)	-20°C to 55°C (-4°F to 131°F) Discharge 0°C to 45°C (32°F to 113°F) Charge
Pack Operating Temperature (Max Cycle Life)	0°C to 30°C (32°F to 86°F) Charge/Discharge
Recommended Tempurature (Air) - With ESS Underground	-73°C to 76°C (-100°F to 170°F)
Recommended Tempurature (Soil)	0°C to 37°C (32°F to 100°F)
Humidity	Up to 100%, condensing, standing water

Safety Specifications

Pack Thermal Runaway (TR) Mitigation	Positive Argon pressure keeps TR in cell casing. Hermetic environment starves initial fire of oxygen
Pack TR Propagation Prevention	Pressurized Argon extinguish flames from ruptures cells
Primary TR Control (Retention)	Pack can retain smoke, gases, chemicals of cell TR, while venting the pressures to prevent pack explosion and deflageration
Secondary TR Control (Retention)	Pack can retain smoke, gases, chemicals of 2nd cell runway, while venting the pressures to prevent pack explosion and deflageration
Tirtiary TR Control (Release)	Pack can release smoke, gases, chemicals of cell TR overboard though a port and customer conduit, to prevent container explosion and deflageration
Proposed Certification	UL9540A, ATEX, UL Class I, Div. 2 for hazardous locations, UN38.3

Dimensions

Height	1.8M
Width	0.6M
Weight	1100kg (2425 lbs)

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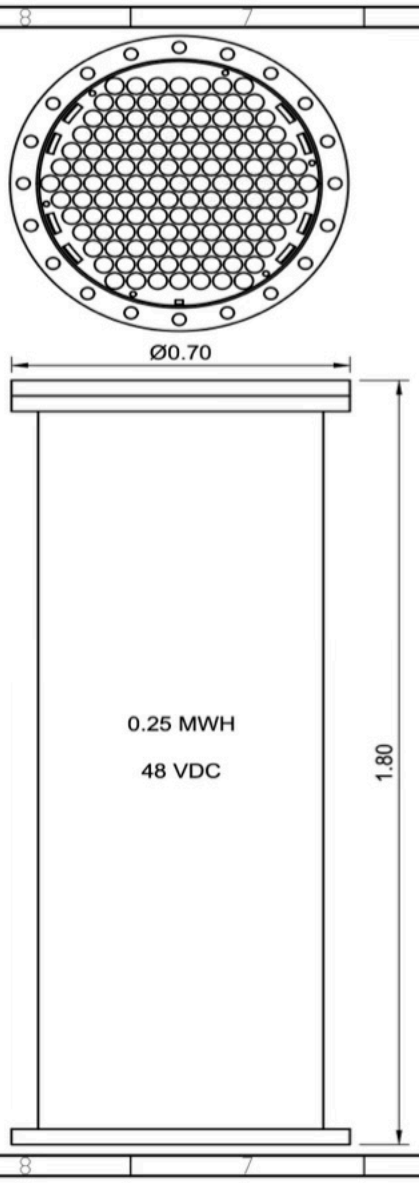

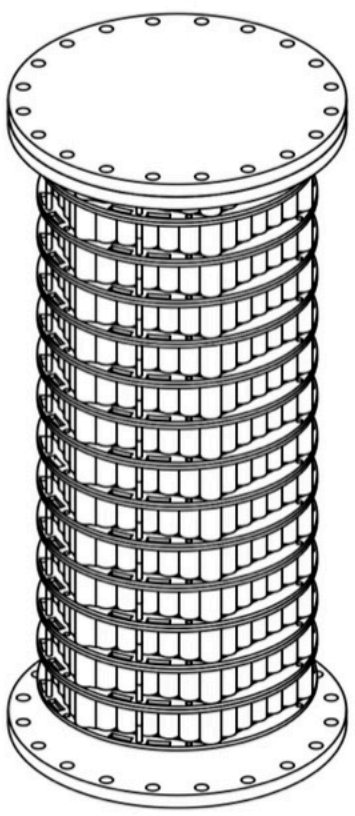

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NOTES:

1. ALL DIMS METERS
2. THERMAL MANAGEMENT: PASSIVE IMMERSION
3. WEIGHT: 1100KG
4. DESIGNED TO SUPPORT OPEN COMPUTE PROJECT: RACK AND POWER WORKING GROUP
5. FUTURE-PROOF, EFFICIENT, AND SIMPLIFIED 48VDC PSU TOPOLOGY FOR NEW-BUILD DATA CENTERS

DRAGON Q ENERGY

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